

Integral Tension Kernels for Compressed Unknot Certificates

Simultaneous multiplier shears, exact flow witnesses,
and a bounded-phase complexity theorem

Research draft for the ProveIt project

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Abstract

We develop a compressed local optimization kernel for the group-certificate branch of the ProveIt unknot recognizer. The existing producer chooses a length-reducing Whitehead direction and then minimizes along its powers. We instead optimize simultaneously over every integral left and right power of a fixed multiplier. The non-multiplier letters cannot cancel, so total cyclic length is an exact weighted integer L^1 tension objective on a graph with only linearly many compressed gap records. Its dual is a capacitated circulation. We give a bit-polynomial solver, a small integer primal–dual witness, exact balanced-graph and forest shortcuts, and an independent verifier.

A direct reduced-image construction is the key representation result: retaining binary-encoded repetition rules, each phase increases grammar size by at most a constant factor, without general compressed free reduction. Thus q strictly shortening phases, including certificates and replay, cost $\text{poly}(\sigma)2^{O(q)}$ for initial encoded size σ . This yields a quasi-polynomial bound when a polynomial-size knot presentation reaches a verified terminal class in $O(\log^2 n)$ phases. Neither that phase bound nor terminal completeness is established for arbitrary diagrams; an infinite family of already automorphically minimal infinite-cyclic presentations exposes the latter obstruction.

The package includes 51 passing tests, 1,200 independently checked flow instances, 360 braid-derived algebraic stages, and reproducible paired measurements. Synthetic multi-run presentations improve, whereas all 21 one-step advantages on the small raw braid corpus disappear after singleton elimination, and the ordinary-stage aggregate shows overhead. These are research-kernel results, not a measured improvement to the complete maintained recognizer.

Delivery status. The mathematical arguments and local code are supplied for independent review. No repository writes, upstream test-suite run, or end-to-end production benchmark were performed. A source-pinned integration gateway is included but was not executed. The claim is not a general quasi-polynomial unknot-recognition algorithm.

Contents

1	Problem, baseline, and precise contribution	4
1.1	Why another compressed group kernel?	4
1.2	Relation to existing complexity results	4
1.3	Results and their scope	4
2	Encoding, contracts, and notation	5
3	The commuting integral shear family	6
3.1	Why retained letters never cancel	7
3.2	The exact objective	7
4	Extracting all gaps without expanding the words	7
5	Integer tension duality and exact witnesses	9
5.1	Reduction to min-cost circulation	9
6	A bit-polynomial circulation implementation	10
6.1	Capacity bits, saturation, and imbalance repair	10
6.2	Correctness and bit complexity	11
7	Exact shortcuts: balanced graphs and median forests	12
7.1	Balanced offset graphs	12
7.2	Forests of parallel-edge bundles	13
8	Direct reduced images and independent replay	14
8.1	Producer construction	14
8.2	A distinct verifier recurrence	15
8.3	Source binding without hash-based mathematics	15
9	A bounded-phase complexity theorem	15
10	Terminal recognition and a genuine completeness obstruction	17
10.1	Pure-power presentations	17
10.2	Why unlimited strict shear descent is not complete	17
11	Separating families and exact phase compression	18
11.1	Unbounded one-step gain over a selected direction	18
11.2	Many independently weighted scales	19
12	Implementation, validation, and measured tradeoffs	19

12.1	What was actually executed	19
12.2	Braid-derived stages, and why the negative result matters	20
12.3	Paired kernel timings	20
12.4	Same-terminal synthetic descent	21
12.5	Certificates and reproducibility artifacts	22
13	Integration into the maintained recognizer	22
13.1	The inspected source contract	22
13.2	Legacy and atomic integration are different projects	22
13.3	Resources, failure, and a conservative scheduler	23
14	Further research questions and concrete next targets	23
14.1	Control changes of multiplier, not only powers	23
14.2	Compress the genuinely frustrated part of the gap graph	23
14.3	Interleave relator moves without losing the phase theorem	23
14.4	Broaden terminal states with explicit verification	24
14.5	Find geometric instances where the local advantage survives	24
14.6	A proof-carrying atomic protocol and formalization	24
15	Conclusion	25
A	Reproduction and file map	25
B	Source provenance and non-execution ledger	26
C	Certificate checks and failure semantics	26

1 Problem, baseline, and precise contribution

1.1 Why another compressed group kernel?

The maintained ProveIt code has already addressed several distinct costs: explicit word expansion, unit Whitehead descent, long relator powers, and compressed overlap discovery. In particular, its power optimizer does not naively repeat an automorphism N times on a word containing a^N . For a selected type-II Whitehead automorphism it computes the exact first minimum by a weighted-median formula on a shared straight-line program. The relevant source is `fast/fastunknot/whitehead_power.py`; the producer chooses its direction earlier, in `compressed_search.py`. The maintained proof is `synthesis/whitehead_powers.tex` [1, 2].

These observations rule out claiming the selected-direction median or simple power batching as a new contribution. The remaining local issue is that a single cut chooses a common exponent for many different oriented generators. Different gaps may want different exponents. Profiling one chosen direction does not optimize that larger family, even if the profile itself is exact.

Our replacement query is: for each multiplier a , choose all integral left/right exponents jointly, then choose the best multiplier. This is a well-defined infinite search space. It admits a finite, bit-polynomial optimization and a separately checkable optimality proof. The resulting operation strictly contains a powered Whitehead move, but is still only a local family inside the full automorphism group.

1.2 Relation to existing complexity results

Roig, Ventura, and Weil give polynomial-time Whitehead minimization in the *expanded* input length and the free-group rank [3]. Their cut-based approach is part of the background here, not a new result of this article. Expanded length can be exponential in compressed description size. Schleimer’s exposition supplies polynomial compressed word and reduction algorithms [4]; the direct constructor below avoids needing those more general routines for this particular family.

Kapovich’s July 2026 preprint establishes fixed-rank compressed primitivity results, including a polynomial rank-two algorithm, and distinguishes testing compressed automorphic minimality from constructing a minimum. Its Open Problem 1.5 explicitly retains the latter distinction [5]. We do not resolve that general minimization problem. We optimize over a single commuting shear family at a time and leave the number of changes of multiplier uncontrolled.

Polynomial min-cost circulation is classical; for example, Goldberg and Tarjan develop a cost-scaling approach [6]. Our capacity-bit implementation is described and proved directly so that its iteration bound does not silently depend on a large capacity’s numeric value. We make no claim that integer tension duality or scaling algorithms themselves are new.

Finally, Lackenby’s July 2026 paper treats incompressible surfaces and hierarchies but states in Section 9 that the implementation detail supplied there is insufficient for a running-time estimate [7]. The present algebraic kernel neither fills in that geometric algorithm nor imports an unverified quasi-polynomial cost assertion from it.

1.3 Results and their scope

The principal results proved here are the following.

- (i) An exact compressed tension formula and a polynomial-size primal–dual certificate for the best

integral shear, uniformly in rank.

- (ii) A direct, literally reduced output grammar of at most constant-factor larger size, together with an independently implemented decorated-word replay.
- (iii) A $\text{poly}(\sigma)2^{O(q)}$ bound for q successful phases, and the corresponding restricted quasi-polynomial recognition corollary.
- (iv) Explicit separating and obstructing families: a one-step gain ratio unbounded in expanded length, and infinite-cyclic presentations on which no strict automorphic shortening is possible.

These are extensions of the inspected repository mechanisms and are fully proved in this draft. Establishing priority over every related algebraic or optimization formulation would require a separate literature investigation. The experimental conclusions are deliberately narrower than the mathematical ones: a stronger local query need not be a faster default recognition stage.

2 Encoding, contracts, and notation

Let $F(X)$ be the free group on $X = \{x_1, \dots, x_r\}$. We represent a signed letter by a nonzero integer x , with $-x$ denoting its inverse. The set of *alive generators* is supplied explicitly; a generator absent from all current relators must not be silently deleted. Such a generator contributes a free factor to the presented group.

The input is a tuple $\mathcal{W} = (w_1, \dots, w_m)$ of literally freely and cyclically reduced words, including possible empty words and duplicate roots. Relator slots are retained. Write

$$N = \sum_{i=1}^m |w_i|, \quad \mathcal{P} = \langle X \mid w_1, \dots, w_m \rangle.$$

For reduced cyclic words, $|w_i| = |w_i|_{\text{cyc}}$. A cyclic rotation of a relator preserves its normal closure, but can change the meaning of a later *positional* overlap record. This distinction is important in Section 13.

Definition 2.1 (Repetition grammar). A repetition grammar is a topologically ordered acyclic list of rules of the forms

$$0 \rightarrow \epsilon, \quad T \rightarrow x, \quad C \rightarrow UV, \quad P \rightarrow U^k,$$

where $k \geq 0$ is written in binary. Empty, zero-power, and unit-power rules are simplified by the supplied implementation. A root list specifies the relators. A binary SLP is the special case without repetition rules.

Repetition rules do not add a stronger compression model for polynomial-time purposes: binary powering replaces U^k with $O(\text{bl}(k))$ concatenation rules. They do matter for the sharper *per-phase* representation bound. Expanding all of them after every phase introduces a bit-length factor that our direct bound intentionally avoids.

A bottom-up pass stores literal length, first letter, last letter, and a flag for free reduction. A concatenation is freely reduced precisely when its children are and its seam does not contain inverse letters. A nontrivial power is freely reduced precisely when its base is and its cyclic seam is reduced. Thus the input contract is checked without expansion. Since each reachable nonempty rule produces a literal subword of a root, reduced roots also ensure reduced reachable subwords.

Let s be the number of nonempty rules in the supplied representation, $G = s + m + r + 2$, and σ its complete binary encoding size, including root indices, exponent encodings, and generator identifiers. A convenient arithmetic bit bound is

$$B = 2 + \lceil \log_2(N + m + 2) \rceil + \lceil \log_2(G + 2) \rceil + \max_{x \in X} \text{bl}(x), \quad (1)$$

where a maximum over an empty set is zero. Products of multiplicities and lengths may need a constant multiple of B bits, which does not change the bounds. Initially $\log(N + 2) = O(\sigma)$: concatenation adds at most one bit along a path, and repetition adds at most the encoded exponent length.

We use ordinary binary integer arithmetic and charge its bit costs. The conservative global bounds permit quadratic-cost multiplication and comparison-based dictionaries. The Python implementation uses dictionaries; even worst-case quadratic collision overhead is absorbed in the stated polynomial global bound. Faster balanced/forest counts stated in comparison operations refer to standard comparison-map implementations, not a claim of constant-time arithmetic on arbitrarily large integers.

3 The commuting integral shear family

Fix a positive generator $a \in X$. For every signed letter x with $|x| \neq a$, choose an independent integer z_x . Define

$$\Phi_{a,z}(a) = a, \quad \Phi_{a,z}(x) = a^{-z-x} x a^{z_x} \quad (|x| \neq a). \quad (2)$$

The formula on inverse letters agrees with inversion, so it defines an endomorphism of the free group.

Lemma 3.1 (Commuting automorphisms). *For fixed a ,*

$$\Phi_{a,z} \Phi_{a,t} = \Phi_{a,z+t}, \quad \Phi_{a,z}^{-1} = \Phi_{a,-z}.$$

In particular every integral potential vector defines an automorphism.

Proof. The multiplier and all its powers are fixed. Substituting the image of x into $a^{-t-x} x a^{t_x}$ adds the left exponents and the right exponents. This gives the first identity on each generator, and the second follows by setting $t = -z$. $\square$

The family includes powers of the repository's type-II Whitehead convention. For $a \in A$, $-a \notin A$, put $z_x = k[x \in A]$ for all other signed letters. Equation (2) is precisely the k th power of that Whitehead map. Conversely, any z factors into at most $2(r-1)$ powered singleton Whitehead maps. If $z_x > 0$, use multiplier a , subset $\{a, x\}$, and power z_x . If $z_x < 0$, use multiplier $-a$, subset $\{-a, x\}$, and power $-z_x$. All factors commute. The factors need not each decrease length.

Allowing signed z makes negative multipliers redundant for the joint query: $\Phi_{-a,z} = \Phi_{a,-z}$. In cyclic length calculations, adding the same constant to every potential is also irrelevant; it changes only a common conjugation. Additional independent gauge shifts can occur in disconnected gap graphs.

3.1 Why retained letters never cancel

For a non-pure cyclic word retain its cyclic sequence of letters different from $a^{\pm 1}$. Between two successive retained letters u, y is a signed run a^e , possibly empty. Set $v = -y$. Applying the shear changes that gap to

$$e + z_u - z_v. \quad (3)$$

Lemma 3.2 (Skeleton invariance). *After applying any integral shear and taking cyclic free reduction, the cyclic sequence of retained non-multiplier letters is unchanged.*

Proof. Suppose a first cancellation of retained letters were to occur. The two letters would have to be inverse, so $y = -u$ and $v = u$. Their intervening exponent in (3) is then exactly e , independently of z . Since the original cyclic word is reduced, this exponent was nonzero. Those letters cannot become adjacent and cancel, a contradiction. The same argument applies at the cyclic seam. A word containing no retained letter is a pure multiplier power and is fixed. $\square$

The proof does not require nonnegative exponents, a common power, small rank, or any special property of a knot presentation. It is the reason a direct reduced-image constructor is possible rather than merely a length estimate.

3.2 The exact objective

Aggregate equal gap records (u, v, e) over every relator, with positive integer multiplicity w_{uve} . Let C_a be the number of retained letters, counted with multiplicity, plus the total lengths of all pure multiplier roots. Then skeleton invariance gives

$$L_a(z) = C_a + \sum_{(u,v,e)} w_{uve} |e + z_u - z_v|. \quad (4)$$

Self-loops $u = v$ are allowed. Their cost is a constant $w|e|$, not a constraint to be discarded. Empty roots contribute nothing, and a duplicate root contributes a duplicate seam and all its multiplicities.

Corollary 3.3 (Domination and idempotence). *The minimum of (4), followed by minimization over positive multipliers a , is no worse in total cyclic length than any selected-direction power step. Once an exact a -shear minimum has been reached, a second a -shear cannot strictly improve it.*

Proof. Every selected-direction power is in the feasible family. For idempotence, the orbit of a point under the fixed- a family is unchanged after applying one member, by the additive composition law. Its minimum has already been attained. $\square$

Idempotence concerns exact local minima, not equality of returned encodings or uniqueness of potentials. Every successful later shear phase must change multiplier, but this fact alone does not bound how often multipliers alternate.

4 Extracting all gaps without expanding the words

For each reachable node U store four values

$$(f_U, \ell_U, p_U, s_U) : \quad \text{first/last retained letter and signed prefix/suffix } a\text{-run.}$$

When U is a pure multiplier word, $f_U = \ell_U = 0$ and $p_U = s_U$ is its signed exponent. For a non-pure word, the prefix and suffix runs lie outside its first and last retained letters.

A terminal supplies this information immediately. For $U = VW$,

$$\begin{aligned} f_U &= f_V \text{ if } f_V \neq 0, \text{ and } f_W \text{ otherwise,} \\ \ell_U &= \ell_W \text{ if } \ell_W \neq 0, \text{ and } \ell_V \text{ otherwise,} \\ p_U &= p_V \text{ if } f_V \neq 0, \text{ and } p_V + p_W \text{ otherwise,} \\ s_U &= s_W \text{ if } \ell_W \neq 0, \text{ and } s_V + s_W \text{ otherwise.} \end{aligned}$$

If both children contain retained letters, this concatenation introduces the single internal record

$$(\ell_V, -f_W, s_V + p_W).$$

For $U = V^k$ with non-pure V , the four endpoint values are unchanged and there are $k - 1$ additional copies of the seam record $(\ell_V, -f_V, s_V + p_V)$. For pure V , multiply its exponent by k .

Initialize a node multiplicity for each root occurrence, including duplicate roots. Traverse reachable rules in reverse topological order. A concatenation passes its multiplicity to both children, even when the children coincide. A repetition passes k times its multiplicity to its base and contributes $k - 1$ times its multiplicity at the seam. Finally, add one cyclic seam for each non-pure root. Retained terminals give C_a through their propagated counts; pure roots contribute their original lengths separately.

Theorem 4.1 (Compressed gap extraction). *The resulting histogram is exactly the multiset of expanded cyclic gaps. There are at most $s + m$ distinct gap events before equal keys are combined. Metadata, reverse multiplicities, and histogram construction require polynomial bit time and $O(s + m)$ integer records, apart from the input. With comparison dictionaries a bound of $O((s + m) \log(G + 2)B^2)$ bit operations is sufficient.*

Proof. Every internal gap of an expanded parse is created either at a concatenation of two non-pure subwords or at one boundary between copies of a repeated non-pure subword. Such a creation belongs to a unique occurrence of a grammar rule. Reverse multiplicities count those occurrences exactly. Gaps not internal to a root are exactly its one cyclic seam. Pure roots have no such gaps, and their length is included separately.

There is at most one event key per concatenation rule, one per repetition rule, and one per root. Counts can be large, but a nonempty node's occurrences are disjoint in the expanded parse for that node, so their number is at most N ; the bounds in (1) cover their binary encodings. Each metadata or multiplicity update uses a bounded number of additions or multiplications on $O(B)$ -bit integers. Dictionary and sorting costs give the stated bound. $\square$

The certificate verifier does not call this reverse propagation routine. It independently builds a bottom-up histogram for each node: concatenate child histograms and add one seam; multiply a repeated child's histogram by k and add $k - 1$ seams. This costs at most $O(s(s + m))$ histogram records in the straightforward implementation. The extra polynomial factor is useful for implementation independence and does not affect the bounded-phase class.

5 Integer tension duality and exact witnesses

Write E for the gap records and w_e for their weights. The nonconstant primal objective is

$$P = \min_{z \in \mathbb{Z}^V} \sum_{e=(u,v,\eta) \in E} w_e |\eta + z_u - z_v|.$$

Here η denotes a gap's offset; the notation separates it from its record index. A dual witness assigns an integer f_e to each record, with

$$|f_e| \leq w_e, \quad \text{div } f(v) = \sum_{e: \text{tail}=v} f_e - \sum_{e: \text{head}=v} f_e = 0. \quad (5)$$

A self-loop contributes once with each sign and hence zero divergence.

Lemma 5.1 (Weak duality with an exact certificate). *For every integer potential z and every feasible f ,*

$$\sum_e w_e |\eta_e + z_{u_e} - z_{v_e}| \geq \sum_e \eta_e f_e. \quad (6)$$

If equality holds, z is a global integer minimum and f is a matching optimal lower-bound witness.

Proof. Termwise, $w|t| \geq ft$ whenever $|f| \leq w$. Summing and collecting potential coefficients gives

$$\sum_e f_e (\eta_e + z_{u_e} - z_{v_e}) = \sum_e \eta_e f_e + \sum_v z_v \text{div } f(v) = \sum_e \eta_e f_e.$$

The right side is a lower bound for every feasible z , so equality proves optimality. $\square$

This is the complete logical check performed by the small dual verifier. It checks integer types, exact record support, capacity bounds, divergence, and equality of the two integer objectives. It does not trust a floating-point solver, an approximate residual, a fingerprint equality, or a claimed optimal cut. In particular, a nonzero optimal objective is perfectly compatible with a zero *primal-dual gap*.

5.1 Reduction to min-cost circulation

For each record $e = (u, v, \eta)$ introduce two directed arcs:

$$u \longrightarrow v \text{ with capacity } w_e \text{ and cost } -\eta, \quad v \longrightarrow u \text{ with capacity } w_e \text{ and cost } \eta.$$

Let their flows be g_e^+, g_e^- and set $f_e = g_e^+ - g_e^-$. Every feasible circulation gives (5). Its cost is $-\sum_e \eta f_e$. Conversely any feasible f is realized by its positive and negative parts, so there is no relaxation gap from this transformation.

For an arc of cost c , use the reduced-cost convention

$$c^\pi(u, v) = c + \pi_u - \pi_v.$$

Suppose a feasible circulation has potentials making all residual reduced costs nonnegative. Put $z_v = -\pi_v$. For the forward arc associated with a gap, its reduced cost is $-\eta + \pi_u - \pi_v = -(\eta + z_u - z_v)$. If the gap value is positive, that forward arc is saturated and the opposing original arc carries no flow; hence $f_e = w_e$. If it is negative, $f_e = -w_e$. If it is zero, every allowed f_e satisfies the equality for that term. Thus the circulation and its potential give equality in (6).

Theorem 5.2 (Integral optimum and bounded potentials). *Every integer gap instance has an integer optimum and an integer dual witness. One can choose the potential of every component so that*

$$|z_v| \leq (V - 1) \max_e |\eta_e|$$

when $V \geq 1$, with the zero instance interpreted separately.

Proof. The solver in Section 6 constructs an integral feasible circulation and residual-feasible potentials. At termination choose distances from an auxiliary source with zero-cost arcs to every vertex. There is no negative residual cycle. A shortest path may be taken simple, so its cost lies between $-(V - 1) \max |\eta|$ and zero. The distance inequalities give nonnegative residual reduced costs. Their negatives are the required integral potentials, and the preceding complementary-slackness argument supplies equality. $\square$

The bound depends on the *bit length* of the offsets, not a search through all integer vectors in this coordinate box. Enumerating the box would undo the compression advantage.

6 A bit-polynomial circulation implementation

The reference implementation is `shear_kernel/flow.py`. This section proves the invariants of the actual algorithm rather than appealing to unit-capacity augmentation. Let A be the number of original directed arcs, V the number of vertices, and

$$\beta = \max_i \text{bl}(U_i), \quad \gamma = 1 + \max_i \text{bl}(c_i),$$

for capacities U_i and costs c_i . Zero-arc instances return immediately. A constant offset in these bit-length definitions is harmless.

6.1 Capacity bits, saturation, and imbalance repair

Initially all partial capacities and all flows are zero, and all vertex potentials are zero. Read capacity bits from most significant to least significant. At a new phase replace each partial capacity and flow by

$$U'_i \leftarrow 2U'_i + \text{next bit of } U_i, \quad g_i \leftarrow 2g_i.$$

Before the new bit, doubling preserves feasibility and the support of every residual arc. The potential therefore still certifies their nonnegative reduced costs.

If a newly available forward residual arc has negative reduced cost, saturate it. The extra residual capacity is at most one: a negative reduced cost meant that its predecessor was saturated, so only the newly read bit can open it. Its reverse residual arc has positive reduced cost. After all such saturations every residual reduced cost is nonnegative, but flow conservation may fail. Let

$$b(v) = \text{inflow}(v) - \text{outflow}(v)$$

be the resulting imbalance. The total positive imbalance is at most A . Self-loop saturations create no imbalance.

To repair conservation, run multisource Dijkstra from all vertices with $b(v) > 0$, on nonnegative residual reduced costs. Choose a reachable deficit vertex t , $b(t) < 0$, of minimum distance D . Let $d(v)$ be its computed distance, with infinity for unreachable vertices. Update

$$\pi_v \leftarrow \pi_v + \min\{d(v), D\}, \quad \min\{\infty, D\} = D. \quad (7)$$

Augment along the shortest path to t by the minimum of its residual capacities, its source surplus, and its target deficit magnitude. Repeat until all imbalances vanish.

Lemma 6.1 (Every repair exists and reduces the imbalance). *If some positive imbalance remains, a positive vertex has a residual path to a negative vertex. Every repair reduces the total positive imbalance by a positive integer. Hence a capacity phase uses at most A repairs.*

Proof. The doubled old circulation is a feasible circulation under the new capacities. Subtract the current capacity-feasible pseudoflow from that circulation. Positive and negative differences are residual arcs. Their divergence corrects the current imbalance, so decomposition into paths and cycles gives a path from some surplus to some deficit. Every path capacity and nonzero endpoint imbalance is integral and positive. The augmentation changes only its two endpoint imbalances and reduces their magnitudes without reversing their signs. Thus the total positive imbalance falls by at least one. $\square$

Lemma 6.2 (Residual-potential invariant). *After (7), all residual reduced costs are nonnegative and each edge on the selected path has zero reduced cost. Augmenting along that path preserves residual nonnegativity.*

Proof. Let $\hat{d}(v) = \min\{d(v), D\}$. For a residual arc with nonnegative reduced cost $c^\pi(u, v)$, shortest-path inequalities imply

$$\hat{d}(v) \leq \hat{d}(u) + c^\pi(u, v).$$

This also holds when a distance is infinite: a reachable vertex cannot point to an unreachable vertex along a residual arc, and the other cases follow from truncation and nonnegativity. The new reduced cost is $c^\pi(u, v) + \hat{d}(u) - \hat{d}(v) \geq 0$. Every prefix distance on the chosen shortest path is at most D , so those edges have equality. Augmentation can only create reverse edges of that path; their reduced cost is also zero. $\square$

At the end of every repair the code recalibrates the potential by Bellman–Ford from a zero-edge auxiliary source. The invariant proves there is no negative residual cycle. Recalibration is not required for correctness, but gives a direct bit-size bound rather than accumulating unconstrained potential shifts across many phases.

6.2 Correctness and bit complexity

Theorem 6.3 (Certified circulation solver). *The capacity-bit algorithm terminates after β phases and at most βA repairs. It returns an integral feasible circulation and integral residual-feasible potentials, and hence an optimum tension witness. A conservative bit-time bound is*

$$O\left((\beta + 1)(V + 1)(A + 1)^2(\beta + \gamma + \log(A + V + 2))\right), \quad (8)$$

with a polynomial space bound. In particular the iteration count is not proportional to a capacity or an offset’s numeric value.

Proof. The initialization satisfies the invariant. Doubling, negative-arc saturation, and the preceding lemmas establish it phase by phase. At the last phase the partial capacities equal the supplied capacities, all imbalances are zero, and residual potentials are feasible. Section 5 then proves optimality and constructs the integer tension solution.

Residual graph construction uses $O(A)$ operations. One Dijkstra run uses $O((A + V) \log(V + 2))$ comparisons and additions. Recalibration uses $O(VA)$ relaxations. The latter bound, with harmless degenerate-instance adjustments, absorbs the former for a graph whose vertices are incident to input arcs. There are at most A repairs per phase and β phases. Saturations and capacity doubling are lower-order scans.

Capacities and flows have $O(\beta)$ bits. Recalibrated potentials have $O(\gamma + \log(V + 2))$ bits because shortest paths are simple. Dijkstra reduced costs and distances obey the same kind of bound. The indicated additions, comparisons, shifts, and residual updates therefore have the bit cost in (8). Computing the final primal and dual objectives adds polynomially many integer products; in the gap application these are absorbed by the common B^2 bound below. $\square$

The proof also explains two tempting incorrect implementations. Repeatedly sending one unit of flow can take exponentially many iterations on a binary capacity. Repeatedly canceling an arbitrary negative cycle does not inherit the repair count above. Neither is the implementation supplied here.

Corollary 6.4 (Complete local shear query). *Given the repetition grammar and alive generators, one can find the best integral shear over every multiplier, construct its reduced image, and produce and verify witnesses for every multiplier in*

$$O((r + 1)^2 G^2 B^2) \tag{9}$$

bit operations, up to harmless logarithmic factors from the chosen dictionary and bit model. The certificate has $O((r + 1)GB)$ bits. The computation is polynomial in compressed input size even when rank is part of the input.

Proof. For one multiplier, $V \leq 2(r - 1)$ and $A \leq 2(s + m)$. Gap multiplicities and offsets have $O(B)$ bits, so substitute these bounds into (8) and solve at most r instances. Histogram construction, objective products, the independent quadratic histogram verifier, and the direct image construction of Section 8 are absorbed in the same conservative polynomial bound. Each multiplier contributes linearly many record and vertex witnesses of $O(B)$ bits. $\square$

We deliberately use a conservative bound rather than asserting that every Python dictionary access or large integer operation costs one step. The main asymptotic point is polynomial dependence on encoded size, not an optimized exponent for this reference solver. The bounds describe completed computations with resource allowances chosen large enough; a configured work cap may instead interrupt the implementation, which returns no recognition conclusion from such an interruption.

7 Exact shortcuts: balanced graphs and median forests

7.1 Balanced offset graphs

A weighted gap graph is *balanced* when there exist integers z such that

$$z_v - z_u = \eta \quad \text{for every record } (u, v, \eta).$$

A depth-first traversal chooses a root potential zero in each component, propagates the required differences, and checks every already visited edge. This also checks parallel-record consistency and self-loop offsets.

Proposition 7.1 (Balanced fast path). *The graph is balanced if and only if the signed sum of offsets around every undirected closed walk is zero. In that case the primal optimum is zero, $L_a^{\min} = C_a$, and $f = 0$ is an exact dual certificate. A spanning-forest traversal and edge checks decide this using linearly many integer operations apart from dictionary access.*

Proof. A potential telescopes around any closed walk, proving necessity. Conversely, assign potentials by path sums along a spanning forest. Every other edge closes a walk, so its zero offset sum makes it consistent with the assignment. The objective is nonnegative and the assignment makes each term zero. The all-zero dual then has the same value. $\square$

Large multiplicities do not affect this test. Thus binary exponents with hundreds of bits need not trigger hundreds of circulation phases when the underlying offset constraints are exactly compatible.

7.2 Forests of parallel-edge bundles

A self-loop contributes a fixed objective $w|\eta|$ and the independent dual value $f = w \operatorname{sgn}(\eta)$. Remove those loops temporarily. Group all remaining records by their unordered endpoint pair. Suppose the resulting simple graph is a forest. Orient each bundle from a chosen endpoint u to v and put $d = z_v - z_u$. Its objective is

$$\sum_j w_j |d - t_j|,$$

where $t_j = \eta_j$ for a record oriented $u \rightarrow v$, and $t_j = -\eta_j$ for a record oriented $v \rightarrow u$.

Theorem 7.2 (Forest median solution). *Choose a lower weighted median d_{uv} of the t_j in every bundle. Because the simple graph is a forest, all equations $z_v - z_u = d_{uv}$ can be satisfied simultaneously. The resulting potential is a global integer optimum. It has a dual certificate obtained independently inside each bundle, and can be found with $O((s + m) \log(G + 2))$ comparison-map operations on the gap data, plus integer arithmetic.*

Proof. Each bundle's weighted absolute-deviation sum is minimized by a weighted median. On a forest its independently optimal differences integrate to a potential, so the sum of all separate lower bounds is attained.

For the dual, orient net flow g_j in the common bundle direction. When $t_j \neq d$, set $g_j = w_j \operatorname{sgn}(t_j - d)$. At $t_j = d$ choose $g_j \in [-w_j, w_j] \cap \mathbb{Z}$ so that $\sum_j g_j = 0$. This is possible because a weighted median has at most half the weight strictly on either side, so the absolute required correction is no greater than the total median weight. Distribute the integer correction greedily within those capacities. Convert g_j back to each record's orientation. Every bundle has zero net divergence at each endpoint, and every term satisfies complementary equality. Restore the self-loop witnesses. Weak duality proves global optimality. $\square$

Every rank-two multiplier problem has at most two port vertices, so this shortcut solves it after self-loops are separated. Higher-rank forest instances are also covered. A graph can fail the balanced test because two parallel records demand incompatible offsets while still be solved immediately by medians. Conversely, a balanced graph with cycles is handled by the first shortcut even though it is not a forest. Only the remaining instances need the general circulation solver.

These shortcuts change the implementation cost, not the optimization problem or the certificate checker. The benchmark's *forced-flow* arm disables both and therefore offers a same-objective control.

8 Direct reduced images and independent replay

A length optimizer alone would not settle the representation problem: generic substitution followed by compressed free reduction might repeatedly multiply grammar size by a polynomial factor. Skeleton invariance yields something more specific and stronger for the present maps.

8.1 Producer construction

For a non-pure node U build a core M_U starting at its first retained letter and ending at its last. The core contains exactly its transformed internal gaps, but not the prefix or suffix multiplier runs. A pure node has empty core. For a terminal different from the multiplier, the core is that terminal.

For $U = VW$ with both children non-pure, set

$$M_U = M_V a^{s_V + p_W + z_{\ell_V} - z_{-f_W}} M_W. \quad (10)$$

If just one child is non-pure, use that child's core; if neither is, use the empty core. For $U = V^k$ with non-pure V , put

$$h = s_V + p_V + z_{\ell_V} - z_{-f_V}, \quad M_U = (M_V a^h)^{k-1} M_V. \quad (11)$$

For a non-pure root R , output

$$M_R a^{s_R + p_R + z_{\ell_R} - z_{-f_R}}. \quad (12)$$

For a pure root, output its unchanged signed multiplier power. A negative power a^h is encoded as a nonnegative repetition of a^{-1} .

Theorem 8.1 (Direct image theorem). *Equations (10)–(12) give a cyclically reduced representative of every $\Phi_{a,z}(w_i)$. They use no general word-equality, prefix, free-reduction, or expansion query. If $G = s + m + r + 2$, the fresh output representation satisfies*

$$G' \leq 8G. \quad (13)$$

The construction takes polynomial bit time, and its output cyclic length is exactly (4).

Proof. Induct over the grammar. Concatenation joins the two retained cores through exactly the gap in (3). A repeated non-pure core has $k - 1$ internal copy seams, giving (11). At a root, moving its initial multiplier run to its end produces (12). Thus every output has the correct cyclic sequence of retained letters and all the exact transformed gap exponents.

Only opposite retained letters could cancel after a gap becomes zero. Skeleton invariance rules this out, including at the last-to-first seam. Hence the displayed output is already freely and cyclically reduced. Counting retained letters and runs gives exactly (4).

For the rule count, a terminal creates at most one terminal. A non-pure concatenation creates at most one run-power and two concatenations, apart from the two shared multiplier-sign terminals. A repetition creates at most one run-power, one block concatenation, one repetition, and one final concatenation. A root creates at most one run-power and one concatenation. Consequently $s' \leq 4s + 2m + 2$ is a sufficient count for the fresh constructor; the looser (13) also covers the fixed root/generator bookkeeping and empty cases. Each exponent calculation uses a bounded number of operations on the bounded integers. $\square$

The representative is not promised to be the same literal cyclic rotation as an unrelated reduction routine would return. The theorem proves the exact cyclic word and normal closure, not equality of all positional encodings.

8.2 A distinct verifier recurrence

The independent replay module `shear_kernel/replay.py` does not call the producer's core constructor or endpoint summaries. It stores a decorated reduced image

$$\Phi(U) = a^{P_U} H_U a^{S_U},$$

where H_U starts and ends in a retained letter; a pure node is represented by its single signed exponent. For a retained terminal x ,

$$(P, H, S) = (-z_{-x}, x, z_x).$$

For two non-pure children, the decorated concatenation is

$$(P_V, H_V a^{S_V+P_W} H_W, S_W).$$

Pure children add to the appropriate outer exponent. For a non-pure repeated child the middle is constructed in the *other* association

$$H_V (a^{S_V+P_V} H_V)^{k-1},$$

with outer exponents P_V, S_V . Finally a root becomes $H_R a^{S_R+P_R}$. This gives the same literal cyclic representative as the producer's formulas, but can produce a different DAG shape.

Each recurrence follows by substitution and multiplier-run combination. The same non-cancellation proof makes it a valid independent replay without compressed free reduction. Its rule count is again bounded by a constant factor. The tests compare both constructors with each other and with literal substitution where expansion is small.

8.3 Source binding without hash-based mathematics

A standalone optimization certificate stores a SHA256 value of the canonical source DAG serialization. This field detects accidental source mixups; it is not a word-equality oracle. The verifier rebuilds the actual source's histograms and checks every primal-dual witness exactly.

For a multi-phase trace, the first source binding is checked. Later producer hashes refer to a DAG shape different from the independent decorated replay, so they are advisory there. Each later certificate is instead checked against the histograms of the *actual independently replayed words*; its potentials define the next exact automorphism image. Thus no hash collision or unverified serialized intermediate word is used to establish the terminal group claim.

9 A bounded-phase complexity theorem

A successful phase means: solve all multiplier queries exactly, choose a minimum with deterministic tie-breaking, and apply it only when total cyclic length strictly decreases. The generator and relator lists remain fixed during the phase sequence. Any preprocessing or interleaved relator operation must be charged separately.

Lemma 9.1 (Bit lengths during monotone phases). *Let the initial total expanded length be N_0 . During a sequence of successful phases, all current root lengths are at most N_0 , all reachable nonempty rule lengths are at most N_0 , and a chosen potential has magnitude $O(rN_0)$. At phase j , all relevant integers have $O(\sigma + j)$ bits, including growing rule indices.*

Proof. Total cyclic length decreases, and all roots and reachable subwords are literal reduced words, so none is longer than N_0 . Gap offsets and multiplicities are bounded by this length, up to root bookkeeping. The bounded-potential theorem gives the factor $O(r)$.

The direct output has no hidden cancellation. Its nonempty constituent cores, runs, and repeated blocks occur as subwords of a root, apart from at most constant unused multiplier terminals; their encoded lengths are therefore bounded as well. Since $\log(N_0 + 2) = O(\sigma)$ and $G_j \leq 8^j G_0$, rule indices add at most $O(j + \log G_0)$ bits. The same argument applies to the independent replay representation. $\square$

Theorem 9.2 (Bounded-phase search and replay). *For initial encoding size σ , at most q successful integral-shear phases, their complete local optimality certificates, and independent replay can be computed in*

$$\text{poly}(\sigma) 2^{O(q)} \quad (14)$$

bit time and space. A final unsuccessful local query and a polynomial-time terminal predicate on the compressed state obey the same bound.

Proof. The direct theorem gives $G_j \leq 8^j G_0$. The rank is unchanged, and the bit lemma gives $B_j = O(\sigma + j)$. Substitute these into the polynomial local bound and sum over $j \leq q$:

$$\sum_{j=0}^q O\left((r+1)^2 G_0^2 64^j (\sigma + j)^2\right) \leq \text{poly}(\sigma + q) 2^{O(q)}.$$

Polynomial factors in q are absorbed into $2^{O(q)}$; when $q = 0$ the bound is simply polynomial in σ . Source serialization, all multiplier witnesses, and the independent histogram and decorated-word replay have the same type of polynomial dependence on G_j and B_j . Summing their costs and output sizes proves (14). $\square$

This is a representation-sensitive theorem. Merely knowing that each individual generic reduction is polynomial would not suffice: recursively applying an unspecified polynomial-size transformation can give a bound like σ^{d^q} . The constant-factor direct rule construction avoids that problem for this sequence of operations.

Corollary 9.3 (Quasi-polynomial bounded-phase class). *Suppose an input with size parameter n yields, with independently checked provenance, a presentation of encoding size $\sigma \leq n^{O(1)}$. If the specified shear descent reaches a sound, complete terminal decision for that input after $q = O(\log^2(n+2))$ successful phases, the algebraic computation and its replay take*

$$2^{O(\log^2(n+2))} = n^{O(\log(n+2))}$$

bit time. With $q = O(\log(n+2))$ the same part is polynomial-time.

This corollary is a theorem about the class satisfying its explicit hypotheses. It does not assert that arbitrary unknot diagrams have such presentations, phase counts, or terminal states. The next section gives one concrete terminal predicate and proves an obstruction to its universality.

10 Terminal recognition and a genuine completeness obstruction

10.1 Pure-power presentations

Suppose every nonempty root is a pure signed power of one generator. For each alive generator x_j let d_j be the gcd of all its nonzero relator exponents in absolute value, or $d_j = 0$ if it has no such relator. Then

$$\mathcal{P} \cong \bigstar_{j=1}^r C_{d_j}, \quad C_0 = \mathbb{Z}, \quad C_1 = \{1\}. \quad (15)$$

A bottom-up uniform-letter test recognizes the roots, and binary Euclidean arithmetic computes the d_j without expanding their powers.

Proposition 10.1 (Complete pure-power terminal test). *The group in (15) is infinite cyclic if and only if exactly one d_j is zero and every other d_j is one. This is decidable in polynomial bit time in the compressed state.*

Proof. Relations involving only one generator present its cyclic factor; the universal property of free products gives (15). If the stated condition holds, the only nontrivial factor is $\mathbb{Z}$. With no nontrivial factor the group is trivial. With one finite nontrivial factor it is finite cyclic. With two or more nontrivial factors, reduced free-product normal forms show that nonidentity elements from distinct factors do not commute, so the group is not $\mathbb{Z}$. These cases exhaust the possibilities. Uniform-letter metadata and gcd arithmetic have polynomial bit cost. $\square$

For a validated classical knot in S^3 , the full knot group is infinite cyclic exactly when the knot is the unknot. To recall the topological input: if the group is $\mathbb{Z}$, the longitude is null-homotopic. Dehn’s lemma gives an embedded compressing disk with that boundary, and an annulus in the tubular neighborhood extends it to a spanning disk for the knot. The converse follows from the solid-torus exterior of the unknot. This is the classical Dehn-lemma/loop-theorem consequence used by the repository’s group verifier [8, 1].

Thus the phase theorem and this terminal predicate give a sound two-sided recognition algorithm *on the class of sourced knot presentations that reach this terminal predicate within the stated phase budget*. The artifact’s standalone CLI deliberately reports only an abstract presentation result: it does not itself prove that a user-supplied presentation is the full group of the claimed diagram. A quotient, an omitted relator, or a mislabeled multi-component link cannot be substituted for that provenance contract.

10.2 Why unlimited strict shear descent is not complete

Theorem 10.2 (Automorphically minimal infinite-cyclic plateau). *For every $N \geq 2$, the presentation*

$$\mathcal{Q}_N = \langle a, b \mid a^N, a^{N+1}, [a, b] \rangle, \quad [a, b] = aba^{-1}b^{-1}, \quad (16)$$

presents $\mathbb{Z}$, has no generator occurring exactly once in any relator, and is already globally minimum in total cyclic length under all free-group automorphisms. Its minimum is $2N + 5$. It is not a pure-power terminal state.

Proof. The first two relations imply $a = 1$, and the commutator then becomes redundant; b is unconstrained. This proves the group is $\mathbb{Z}$. The occurrence assertion is immediate for $N \geq 2$.

For any automorphism ψ , a nontrivial element $\psi(a)$ has positive cyclic length. Cyclic lengths of its powers therefore give

$$\left| \psi(a^N) \right|_{\text{cyc}} + \left| \psi(a^{N+1}) \right|_{\text{cyc}} = (2N + 1) |\psi(a)|_{\text{cyc}} \geq 2N + 1.$$

The element $\psi([a, b])$ is nontrivial and belongs to the commutator subgroup. A nontrivial cyclically reduced word with zero abelianization has length at least four: length one or three cannot have every exponent sum zero, and a length-two zero-sum word cancels. Hence its cyclic length is at least four. The initial tuple has total length $2N + 5$, so it attains this universal lower bound. Its commutator root is not a pure power. $\square$

This is a completeness obstruction to *strict automorphic shortening alone*, not a lower bound for unknot recognition. It does not obstruct relator combination, quotient deletion, or Tietze elimination. Indeed the repository's existing relator-power mechanisms are precisely relevant to such an example. It also is not a family of hard knot diagrams: it is a transparent family of supplied presentations of an infinite-cyclic group.

At $N = 2^{500}$ the implementation confirms the exact local minimum and returns an inconclusive stall, not a knottedness verdict. The example shows why proving only a phase bound would still not establish terminal completeness. One must understand how non-automorphic presentation operations interact with the representation and search bounds.

11 Separating families and exact phase compression

11.1 Unbounded one-step gain over a selected direction

Proposition 11.1 (A selected median can choose the wrong scale). *For $N \geq 3$, consider the tuple*

$$\mathcal{W}_N = ((ba^N)^2, (ca)^3), \quad L_0 = 2N + 8.$$

The best unit Whitehead direction has gain five. Exact line minimization along any such best direction uses exponent one and leaves length $2N + 3$. An integral a -shear reaches length five in one step. Thus the ratio of one-step gains is $(2N + 3)/5$, unbounded in expanded length.

Proof. For multiplier a , the gap objective is

$$5 + 2|N + z_b - z_{-b}| + 3|1 + z_c - z_{-c}|.$$

A unit direction can decrease each positive gap by at most one; choosing both differences -1 realizes gain $2 + 3 = 5$. A unit map with multiplier b can remove at most the two occurrences of b , and one with multiplier c can remove at most the three occurrences of c . Words without that multiplier cannot shorten because all their corresponding initial gaps are zero. Hence a , or its equivalent signed direction, is best.

Along a best direction the length is $5 + 2|N - k| + 3|1 - k|$. Its first minimum is $k = 1$, the lower weighted median. Independently setting the two differences to $-N$ and -1 gives length five. Finally every automorphic image of these two nontrivial powers has length at least two plus three, so five is even a minimum over the full automorphism group for this tuple. $\square$

The example does *not* give an exponential running-time advantage over the already batched repository method. A subsequent selected-direction power can remove the remaining long run; this simple fixture separates one step from two, not one step from N unbatched steps. Its purpose is to establish strictly stronger local optimization without mischaracterizing the baseline.

11.2 Many independently weighted scales

For $k, h \geq 1$ put

$$N_j = j2^h, \quad w_j = 2^{h+2(k-j+1)}, \quad u_j = b_j a^{N_j} \quad (1 \leq j \leq k), \quad (17)$$

and consider

$$\mathcal{S}_{k,h} = \langle a, b_1, \dots, b_k \mid u_j^{w_j}, u_j^{w_j+1} \ (1 \leq j \leq k) \rangle. \quad (18)$$

There are only $O(k)$ repetition rules, with encoding size $O(k(h + k + \log(k + 2)))$. The combined weight at the j th scale is $v_j = 2w_j + 1$, and these weights satisfy

$$v_j > \sum_{\ell > j} v_\ell.$$

Theorem 11.2 (One phase versus k selected-line phases). *The group (18) is infinite cyclic. Joint integral-shear optimization reaches a pure-power terminal state in one phase, at the globally smallest possible total cyclic length $\sum_j v_j$. A greedy best-unit direction followed by an eager first line minimum takes exactly k positive phases, with multiplier a preferred in ties.*

Proof. Setting $z_{b_j} - z_{-b_j} = -N_j$ maps every u_j to the cyclic word b_j . The pair of resulting exponents $w_j, w_j + 1$ has gcd one, so all b_j are killed and a is free. Each original nontrivial power has every automorphic cyclic image at least as long as its exponent; consequently $\sum_j (2w_j + 1)$ is also the global automorphic lower bound, attained here.

At an intermediate selected-line state, each unfinished scale has a positive a -run, whereas finished scales have zero run. The best unit a -direction reduces every unfinished gap by one and changes no finished gap, gaining the sum of the unfinished v_j . A unit direction with any other multiplier b_j can remove at most its own v_j occurrences; all other roots have no b_j and cannot shorten. If its scale is finished, its own roots are pure b_j powers and are fixed. Thus a is a best unit multiplier, strictly so when at least two scales remain.

Along the selected direction, the least unfinished run has weight greater than all later weights combined. Its length is the unique weighted-median scale. The first line minimum therefore removes exactly this least run and subtracts the same amount from all later runs. Repeating removes the scales in increasing order, one per phase. There are exactly k of them. $\square$

This is an exact, rank-dependent phase compression, not a hard-diagram construction. It provides a useful capacity and scaling test for the kernel and explains why the synthetic benchmark can show larger gains than the ordinary braid-stage corpus. The theorem also holds for many other increasing run sequences and superincreasing combined weights.

12 Implementation, validation, and measured tradeoffs

12.1 What was actually executed

The delivered implementation is standard-library Python, executed on CPython 3.13.5 under Linux. It contains separate modules for literal repetition grammars, reverse gap extraction, min-cost circulation, forest medians, joint optimization, decorated replay, terminal arithmetic, a CLI, and a proposed WordArena bridge. There is no dependency on an LP package or floating-point arithmetic.

The retained unit-test log records 51 passing tests. Tests exercise malformed rules and certificates, reducedness preconditions, signed multipliers, inverse-neighbor gaps, duplicate roots, huge powers, small exhaustive potential searches, random exact witnesses, literal substitution and inverse checks, forest-versus-flow comparisons, and multi-phase independent replay. The gateway-shaped fixture is not a substitute for executing the actual upstream module.

The separate deterministic audit uses seed 20261008 and records:

Audit item	Completed count
General circulation instances, fast paths disabled	1,200
Capacity phases across those instances	37,563
Imbalance repairs across those instances	12,119
Connected braid closures retained	180
Raw and post-elimination algebraic stages	360
Individual relator-image comparisons	1,012
Serialized large abstract-presentation traces	3
Failures in the completed audit	0

The circulation audit includes hundreds-bit capacities and offsets. Every result is checked by the independent primal–dual equality; small cases in the unit tests are also compared with exhaustive bounded-potential searches. There are an additional 120 randomized forest-versus-circulation comparisons in the unit tests. Such tests supplement, rather than replace, the solver proof and the no-cancellation theorem.

12.2 Braid-derived stages, and why the negative result matters

For a braid generator σ_i , the independent Artin convention sends

$$(x_i, x_{i+1}) \mapsto (x_i x_{i+1} x_i^{-1}, x_i),$$

with its explicitly implemented inverse. Composing the braid word gives closure relators $\beta(x_i)x_i^{-1}$. The induced permutation checks that the closure has one component. Inverse-pair and braid-relation tests check the convention. This constructs a separate source corpus; it does not call the production PD-to-Wirtinger constructor.

The audit retains 180 connected closures on two through five strands with one through twelve input crossings, obtained in 627 generation attempts. For each it checks the raw presentation and the presentation after exact singleton Tietze elimination. Both tuples, all elimination records, and length results are retained. These two stages of the same input are not statistically independent samples.

A stronger joint query is never worse in resulting length, as the theorem predicts. It is strictly better on 21 raw stages, but on *zero* post-singleton-elimination stages. For example, the four-braid $(-3, 1, 3, 2, 3)$ has raw length 12; a selected-line step leaves 10, whereas a joint step leaves 6. The singleton prelude already eliminates all relators on this example. Reporting its raw gain as a production recognition speedup would therefore be misleading.

12.3 Paired kernel timings

`experiments/benchmark.py` uses one excluded warm-up for each arm, then five measured rounds with shuffled arm order for each case. Inputs are preconstructed and immutable; output construction

is timed. Joint arms include the independent optimality witness check. The selected-line arm independently checks its output-length formula, but does not produce a certificate of global joint optimality. It is a mathematical same-harness baseline, not the repository’s adaptive trigger policy or its measured runtime.

The two joint arms have the *same optimization contract*. The structured arm uses balanced and forest shortcuts, and the forced-flow arm disables both. The table sums per-case medians in milliseconds; it is not a single wall-clock run of the complete recognizer.

Stage corpus	Strict length gains	Selected line	Joint structured	Joint flow
Raw presentations	21/180	33.717	107.619	114.320
After singleton elimination	0/180	12.643	27.723	35.657

There are 5,400 measured calls in this part. Across both stages the sums are 46.360 ms for the selected-line baseline, 135.342 ms for structured joint optimization, and 149.977 ms for forced flow. Thus the more powerful query has substantial overhead on these ordinary small inputs. The shortcuts reduce that overhead, but the corpus does not support replacing the default policy.

12.4 Same-terminal synthetic descent

The synthetic experiment measures *complete descent to the same pure-power infinite-cyclic terminal*, using (18). Both methods check their output data; the joint method also supplies its stronger local optimality witnesses. No measured baseline in this table is capped or incomplete. All samples are retained, including execution order.

k	h	Line (ms)	Joint (ms)	Flow (ms)	Phases line/joint	Line/joint
3	20	1.139	0.583	1.796	3/1	1.95
6	20	7.114	1.608	6.345	6/1	4.42
12	20	62.703	5.116	25.118	12/1	12.26
3	100	1.012	0.589	5.190	3/1	1.72
6	100	6.728	1.700	18.019	6/1	3.96
3	500	1.055	0.672	22.107	3/1	1.57
6	500	7.033	1.691	77.832	6/1	4.16

There are 105 measured calls. The observed phase counts agree with the exact theorem: k versus one. At $(k, h) = (12, 20)$, medians are 62.703 ms and 5.116 ms, approximately 12.26 times apart. At $(6, 500)$ they are 7.033 ms and 1.691 ms, approximately 4.16 times apart. These are completed synthetic-presentation measurements, not ratios against censored runs and not unknot-diagram benchmarks.

For the identical joint query at $(6, 500)$, disabling the shortcuts raises the median from 1.691 ms to 77.832 ms. This isolates the cost of processing many capacity bits unnecessarily on structured graphs. The large integer sizes remain encoded throughout; no 2^{500} -letter materialization occurs.

Five repetitions do not justify fine-grained confidence claims. Host load, interpreter effects, different source grammars, and verification contracts can change timings. The robust evidence is the exact phase difference and the clearly separated completed measurements on these fixtures; no universal speedup is inferred.

12.5 Certificates and reproducibility artifacts

The three supplied JSON traces use exponent bit parameters 20, 100, and 500 on six-branch infinite-cyclic presentations with consecutive pure-power exponents. Each trace is independently replayed from its original compressed presentation. The CLI prints *abstract presentation only; no knot verdict*. The gap witnesses, raw timing samples, and source corpus can be inspected without a production installation.

The local test suite and the audit were executed. The complete maintained suite, real upstream gateway, PD-based pipeline benchmark, and any Rust or Lean integration were not executed. Their absence is recorded in `integration/STATUS.json`, not concealed by reporting the repository’s own historical passing-test counts as ours.

13 Integration into the maintained recognizer

13.1 The inspected source contract

The inspected Python source snapshot is commit

81a387aada56bb180649a98c21c078cb89efcfde.

The source-manifest appendix records exact blob identities and which ranges were read. The maintained selected-direction implementation uses t/c SLP rules, separate free/cyclic reduction, a min-cut direction, and an adaptive decision about whether to profile the selected power. Our baseline timing does not emulate that complete scheduler.

The delivery is additive. A suitable initial destination is `Topology/UnknotRecognition/research/integral_shears_20261008/`. No production file should be replaced automatically on the evidence here.

13.2 Legacy and atomic integration are different projects

The bridge supplies `prepare_step`, which imports the arena’s current roots and prepares a checked joint solution without replacing the source-root list. It can translate the solution into the existing `whitehead/whitehead_power` records using singleton factors.

For legacy certificates, `replay_in_arena` applies those factors using the host’s own substitution and cyclic reduction. This preserves the host’s representative convention. Its returned roots and move records must be committed together only after complete success and independent full-source replay. This route does not inherit the direct rule-growth bound, and its intermediate steps can increase length.

In contrast, `install_prepared` imports the direct core output. It is reserved for a *new atomic-shear protocol*. Installing these roots while recording only legacy positional moves can make a later overlap certificate refer to the wrong positions. A new protocol must specify the cyclic root convention, signed potential map, source identities, replay recurrence, resource handling, and schema version explicitly.

The source-pinned gateway checks blob hashes, imports the real WordArena, compares legacy replay with the joint result on small inputs, and checks the existing power-profile formula. It is supplied for maintainer execution and is explicitly marked **not executed** in this delivery. Passing it would still not be an end-to-end integration benchmark.

13.3 Resources, failure, and a conservative scheduler

The circulation **Budget** counts cooperative flow work. It is not a whole-operation wall-time or memory limit. Histogram extraction, structured shortcuts, serialization, large integer operations, and host allocations need a shared outer deadline and allocation accounting before service use. Arena allocations made before a failed tentative step are not rolled back; a scratch arena or explicit accounting of abandoned nodes is required.

A budget failure, a non-pure terminal, or a shear stall is inconclusive. The existing complete exact backend must remain available. The current small-corpus evidence supports an opt-in probe for highly compressed or multi-scale states, not a new default. Even a perfect local optimizer can choose a different long-term path than a less aggressive heuristic.

A production comparison should measure original PD construction, unchanged filters, search, independent source replay, resource exhaustion, and the final verdict in paired shuffled runs. It should retain no-change controls and negative results. A smaller algebraic state or a better local length objective is not by itself a completed recognition speedup.

14 Further research questions and concrete next targets

14.1 Control changes of multiplier, not only powers

The fixed-multiplier orbit is solved exactly and is idempotent. The next structural question is whether a natural class of knot-derived presentations admits a bound on the number of successful *changes* of multiplier. A useful target is $O(\log^2 n)$ phases with a polynomial-size certified prelude. An initial theorem could restrict rank, a graph decomposition, or a geometric presentation parameter. Any claimed bound must permit arbitrarily large encoded gap exponents rather than reducing to a bound in expanded length.

A concrete experimental target is to retain the whole sequence of optimizing multipliers, dual circulations, state sizes, and length gains on adversarial rather than only random presentations. Seek either a monotone potential controlling changes of multiplier or an explicit long alternating family. The mere fact that each step shortens length is not enough.

14.2 Compress the genuinely frustrated part of the gap graph

Balanced graphs and forests are solved without general flow. In a general graph, bridges connect otherwise independent blocks and should be removable by compatible one-variable optimizations. Can one formulate a certified block decomposition whose expensive part depends only on the size of the cyclic, offset-inconsistent core? A useful theorem would bound the cost by a small frustration parameter plus near-linear processing of the remaining records, still with binary capacities.

A further target is incremental optimization: after a local relator operation, reuse an existing dual circulation and repair only the changed gap records. The reused witness is valuable because exact primal–dual equality can certify success without trusting the incremental search procedure.

14.3 Interleave relator moves without losing the phase theorem

The plateau family proves that automorphisms alone are insufficient for the pure-power terminal strategy. The repository already has useful compressed relator-power and overlap operations. The

key next theorem is not merely soundness of one more move; it is a combined representation bound for a sequence mixing shear phases, generator elimination, quotient deletion, and partial relator replacement.

One precise goal is an additive or constant-factor grammar bound for a certified round consisting of one shear phase and a maximal batch of a restricted relator operation. A phase potential must then count both rounds and their effect on rank and root identities. An unrestricted appeal to a polynomial compressed reduction primitive does not establish the desired end-to-end bound.

14.4 Broaden terminal states with explicit verification

Pure powers are a deliberately transparent terminal class. A more useful terminal could recognize signed generator identifications, free-product splittings, or a bounded number of residual mixed relators after gcd elimination. Each extension should return its own replayable proof that the full presented group is, or is not, infinite cyclic.

On sourced knot inputs, topology may rule out some abstract free-product possibilities. Such facts should sharpen a verified decision rule rather than justify silently deleting relations or inferring a knot verdict from an unsourced group. A complete diagram-to-presentation chain remains essential.

14.5 Find geometric instances where the local advantage survives

All observed raw-stage gains in the current small braid corpus are preempted by singleton elimination. The next meaningful corpus should therefore condition on surviving the existing cheap stages, include larger source braids and difficult PD diagrams, and retain the source construction and all censored runs. This is a concrete test of relevance, not an invitation to inflate arbitrary group presentations and call them difficult knots.

The synthetic families establish encoded capacity and phase compression. A separate mathematical target is an explicit diagram family for which the maintained preprocessing produces a comparable multi-scale tension structure with polynomial overhead. Such a family would connect the local theorem to an actual recognition-performance class.

14.6 A proof-carrying atomic protocol and formalization

The direct constructor is where the strongest representation theorem lives. Formalize its signed-letter, seam, repetition, and normal-closure semantics, then specify an atomic certificate version that does not confuse cyclic representatives with positional records. The independent decorated recurrence provides a small, tractable formalization target.

The arithmetic certificate is another suitable target: formalize the identity

$$\sum_e f_e(z_{u_e} - z_{v_e}) = \sum_v z_v \operatorname{div} f(v)$$

and deduce integer optimality from primal–dual equality. This verification lemma is considerably smaller than formalizing the optimizer. Afterwards, formalizing the scaling invariant and bounded-phase recurrence would connect executable witnesses to a machine-checked cost argument.

15 Conclusion

This work replaces a selected powered Whitehead direction with an exact optimization over a whole commuting family of integral multiplier shears. The important local fact is that non-multiplier letters never cancel; the important algorithmic fact is that the resulting integer tension has an exact circulation witness; the important representation fact is that the reduced output is constructed with constant-factor rule growth.

Together these give an unconditional polynomial compressed local query and a $\text{poly}(\sigma)2^{O(q)}$ theorem for bounded numbers of successful phases. They do not supply the missing geometric phase bound or a complete global presentation strategy. The explicit plateau and the negative post-preprocessing benchmark result make those boundaries concrete. The delivered code, proofs, independent checks, and integration safeguards provide a basis for the next step without upgrading a local improvement into an unsupported general quasi-polynomial claim.

A Reproduction and file map

From the unpacked directory, run:

```
python -B -m unittest discover -s tests -v
python -B experiments/audit.py
python -B experiments/benchmark.py
python -B experiments/make_tables.py
python -B -m shear_kernel verify certificates/star_Z_500.json
sh build.sh
```

The CLI also generates an independently replayed example:

```
python -B -m shear_kernel demo --bits 500 --branches 6 \
  --output demo.json
```

The result is an abstract-presentation certificate, not a diagram certificate. To run the unexecuted real-source gateway in a full checkout:

```
python -B integration/check_upstream.py \
  --fast-root /path/to/ProveIt/Topology/UnknotRecognition/fast \
  --output results/upstream_gateway.json
```

File or directory	Purpose
article.tex, article.pdf	This source and compiled article.
shear_kernel/slp.py	Literal repetition grammar and reducedness checks.
shear_kernel/histogram.py	Producer gap extraction, independent histogram recurrence, direct core output.
shear_kernel/flow.py	Capacity-bit circulation, balanced shortcut, primal–dual verifier.
shear_kernel/forest.py	Exact parallel-bundle median forests.
shear_kernel/optimize.py	All-multiplier optimization, certificates, legacy factorization.

File or directory	Purpose
<code>shear_kernel/replay.py</code>	Independent decorated-word image recurrence.
<code>shear_kernel/presentation.py</code>	Bounded descent and pure-power arithmetic terminal.
<code>shear_kernel/reference.py</code>	Literal checks, Artin presentations, selected-line baseline.
<code>shear_kernel/adaptor.py</code>	Proposed, non-default WordArena bridge.
<code>tests/</code>	Executed local unit tests.
<code>experiments/</code>	Deterministic audit, paired benchmarks, table generator.
<code>results/</code>	Test/build logs, raw samples, audit summaries, source corpus.
<code>certificates/</code>	Three large compressed abstract-presentation traces.
<code>integration/</code>	Source pin, execution status, instructions, unexecuted gateway.
SHA256SUMS	Integrity hashes for the delivered files other than the manifest itself.

No third-party article PDF or upstream implementation source is redistributed. The package is licensed MIT-0. The Python source and numerical output are inspectable without compiling the PDF.

B Source provenance and non-execution ledger

The following blob identities were returned during GitHub source inspection. A full checkout was not available to execute in the working environment. The first four files below were explicitly fetched at the pinned commit; the maintained theory file was fetched from `main` during the same inspection.

File and inspected range	Git blob identifier
<code>whitehead_power.py</code> , complete	d235def0cf042e9e6407c89512b023127d811ec4
<code>compressed_search.py</code> , 1–110	ff33437267d37bc874b00cb4e087f083b94e1a7b
<code>group_certificate.py</code> , 1–150	922510eda5b7e6731de43e639c7dafcb47a56361
<code>compressed_words.py</code> , 1–110	bd68418e226813a7fa50dc87405c214970ddf50e
<code>synthesis/whitehead_powers.tex</code> , complete	4948b50375b641aa2389782b0d4589bd75bd0156

The machine-readable ledger distinguishes executed local tests from unexecuted integration. In particular, no claim is made that the new `adaptor.py` has passed the actual maintained suite, that the current recognizer has changed defaults, or that repository-reported historical test counts were rerun here. The source-pinned gateway validates file blobs before its limited cross-check, and refuses changed sources unless explicitly overridden after inspection.

C Certificate checks and failure semantics

An optimization certificate contains the alive generator list, initial and optimal total lengths, selected multiplier, a source binding, and one solution record per positive multiplier. Every solution record contains integer potentials and signed bounded flows on the reconstructed gap records. Omitting a multiplier prevents certification of the all-multiplier minimum. Duplicate witness entries, non-integer

objective data, invalid supports, capacity violations, nonzero divergence, and nonzero primal–dual gap are rejected. Solver diagnostics such as repair counts are not mathematical proof premises.

The standalone verifier reconstructs the source histogram independently and checks all witnesses. A trace verifier then uses the separate decorated-word recurrence for each selected automorphism, preserving generator and relator identities. A terminal positive or negative group result is accepted only when recomputation of the terminal predicate agrees. An inconclusive record certifies no knot verdict, regardless of whether its diagnostic reason was a stall or a phase cap.

This is research code, not a hardened remote certificate service. External callers must enforce whole-input, integer-size, elapsed-time, and allocation budgets. Those requirements are implementation-safety obligations distinct from the asymptotic theorem for valid encoded inputs.

References

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- [3] Abdó Roig, Enric Ventura, and Pascal Weil, *On the complexity of the Whitehead minimization problem*, International Journal of Algebra and Computation **17** (2007), 1611–1634. <https://arxiv.org/abs/math/0608779>.
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